Day 1: IPv4 Addressing and Subnet Masks

Level 1, Month 1, Networking foundations

What I learned

- An IPv4 address is 32 bits, written as 4 numbers between 0 and 255. Each number is 8 bits.
- Converting a number to binary with the table `128 64 32 16 8 4 2 1`.
- The mask `/24` means 24 bits for the network and 8 bits for the host. It equals `255.255.255.0`.
- Network address: all host bits are 0.
- Broadcast address: all host bits are 1.
- Valid hosts = 2^(host bits) - 2.

Lab: reading my own machine

From `ipconfig /all` on my computer:

- IPv4 address: `192.168.84.1`
- Subnet mask: `255.255.255.0` (`/24`)

- This looks like a VMware virtual adapter, not my home network.

My analysis:

Item	Value
Network address	`192.168.84.0`
Broadcast address	`192.168.84.255`
First valid host	`192.168.84.1`
Last valid host	`192.168.84.254`
Valid hosts	254

Error log

Mistake	Correct answer	Why
Typed `ipconfing`	`ipconfig`	Typo. The command only works with its exact name.
First valid host `.0`	`.1`	`.0` is the network address

Mistake	Correct answer	Why
		and is reserved.
Network address `192.168.84`	`192.168.84.0`	An address must have 4 numbers.
Broadcast address `1`	`192.168.84.255`	Mixed up broadcast with the host number.
Hosts in `/24`: 253	254	I started from 255 instead of 256.
Hosts in `/25`: 128	126	I forgot to subtract the network and broadcast addresses.

Main lesson: I forgot to subtract the 2 reserved addresses twice. Rule to remember: **2^(host bits) - 2**.

Anki cards created (6)

DNS = 53, HTTP = 80, HTTPS = 443, SSH = 22, DHCP = 67/68, and the valid-hosts rule.

Next session

5 new subnetting problems (network, broadcast, hosts), then start analysing a pcap file.